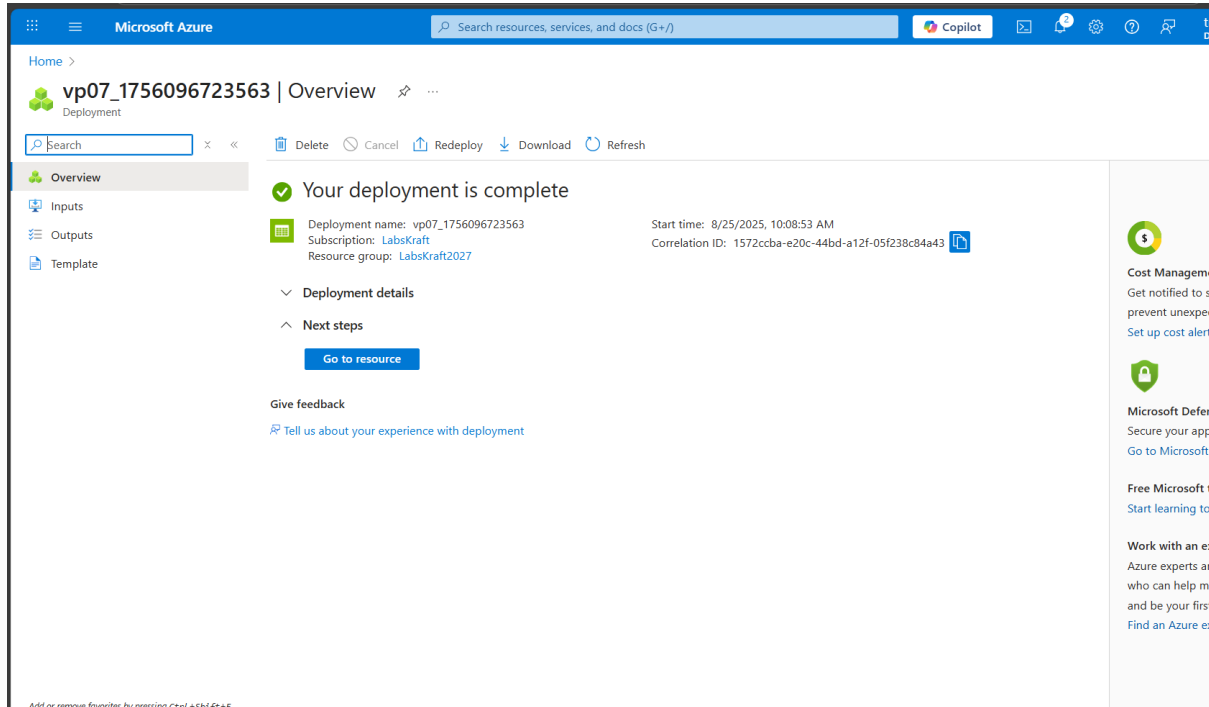
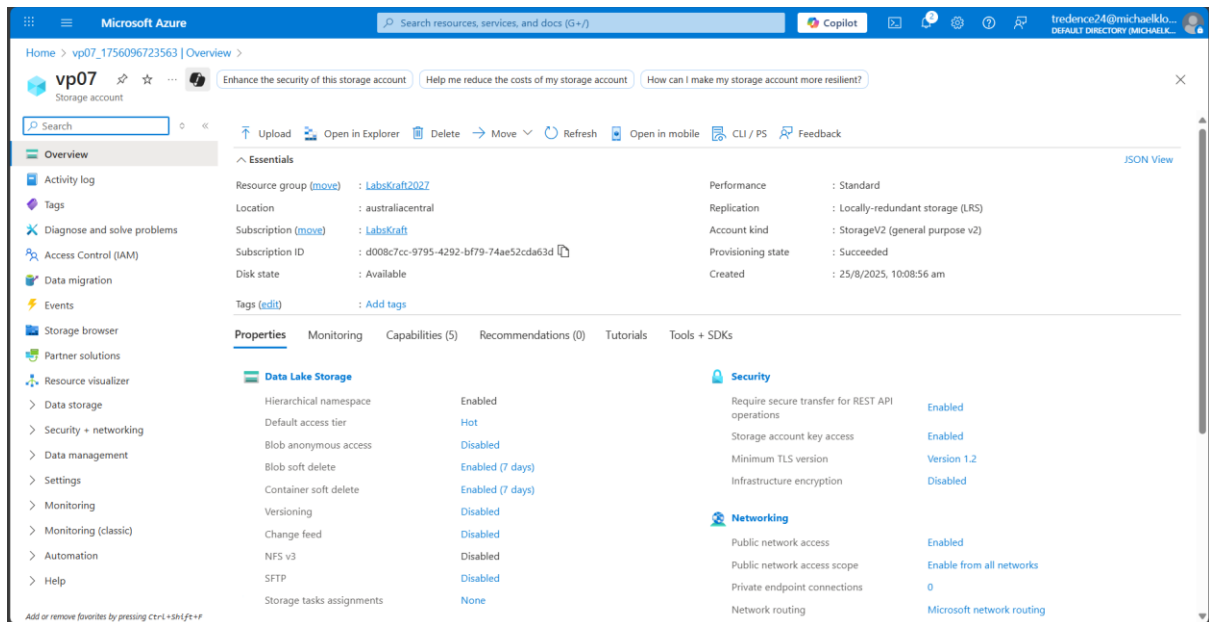


Capstone Project Documentation

1. Create new storage account



2. Create a new container



Microsoft Azure

Search resources, services, and docs (G+/I)

Copilot

Home > vp07_1756096723563 | Overview > vp07

vp07 | Containers ☆ ...

Storage account

Search

+ Add container ↑ Upload ↻ Refresh 🗑 Delete 🔒 Change access level ↺ Restore containers Edit columns

Search containers by prefix

Only show active

Showing all 2 items

☐	Name	Last modified	Anonymous access level
☐	📁 \$logs	25/8/2025, 10:09:38 am	Private
☐	📁 raw	25/8/2025, 10:15:03 am	Private

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Partner solutions

Resource visualizer

Data storage

Containers

File shares

Queues

Tables

Security + networking

Data management

Settings

Monitor

Add or remove favorites by pressing Ctrl+Shift+F

3. Upload blob in raw container

Microsoft Azure

Search resources, services, and docs (G+/I)

Copilot

Home > vp07_1756096723563 | Overview > vp07 | Containers >

raw ...

Container

Search

+ Add Directory ↑ Upload ↻ Refresh 🗑 Delete 📄 Copy 📄 Paste 🔄 Rename 🔒 Acquire lease 🔒 Break lease Edit columns

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

raw

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Only show active objects

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	📄 retail_jot_data.csv	25/8/2025, 10:15:39 am	Hot (Inferred)	Block blob	130.76 KiB	Available

Add or remove favorites by pressing Ctrl+Shift+F

5. Role assignment

Select managed identities



 Some results might be hidden due to your ABAC condition.

Subscription *

LabsKraft



Managed identity

Access Connector for Azure Databricks (14)



 No results were found.



Selected members:



vp07

/subscriptions/d008c7cc-9795-4292-bf79-74ae52cda63d/resourceGroups/... [Remove](#)

Select

Close

 [Feedback](#)

6. Create Synapse workspace

The screenshot shows the 'Overview' page for a Synapse Analytics deployment. The title bar indicates the deployment name 'Microsoft.Azure.SynapseAnalytics-20250825103432'. The left sidebar contains navigation links: Overview, Inputs, Outputs, and Template. The main content area displays a green checkmark and the message 'Your deployment is complete'. Below this, deployment details are listed: Deployment name, Subscription (LabsKraft), Resource group (LabsKraft2027), Start time (8/25/2025, 10:36:42 AM), and Correlation ID. A 'Next steps' section includes a 'Go to resource group' button. A 'Give feedback' section with a feedback icon and text is also present. The right sidebar contains several informational cards: 'Cost management' with a link to 'Set up cost alerts', 'Microsoft Defender for Cloud' with a link to 'Go to Microsoft Defender for Cloud', 'Free Microsoft tutorials' with a link to 'Start learning today', and 'Work with an expert' with a link to 'Find an Azure expert'. At the bottom left, a small note reads 'Add or remove favorites by pressing Ctrl+Shift+F'.

7. Create Catalog

The screenshot displays the 'Catalog Explorer' interface in Databricks, specifically the 'External Locations' view. The left sidebar lists various workspace components: New, Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie, Alerts, Query History, SQL Warehouses, Data Engineering, Job Runs, Data Ingestion, AI/ML, Playground, Experiments, Features, and Models. The main panel shows the configuration for an external location named 'vp07_capstone'. The 'Overview' tab is active, displaying a 'Description' section with an 'Add description' button. Below this, a table lists configuration details: Credential (vp07_capstone_azuremanagedidentity_1756099103065), URL (abfss://root@dbstoragebpzrdv43bl4.dfs.core.windows.net/780045277964770), and Limit to read-only use (Disabled). On the right, the 'About this external location' section shows the Owner as 'Tredence LabsKraft24' and the Fallback mode as 'Off'. A 'Test connection' button is located in the top right corner of the main panel.

Code for Batch Processing

```
%sql
USE CATALOG `vp07_capstone`;
CREATE SCHEMA IF NOT EXISTS vp_schema;
```

```
from dlt import table
import pyspark.sql.functions as F
from pyspark.sql.types import StructType, StructField, StringType, DoubleType,
IntegerType, TimestampType
# :white_check_mark: Define schema for retail IoT CSV
iot_schema = StructType([
    StructField("OrderID", StringType(), True),
    StructField("CustomerID", StringType(), True),
    StructField("Region", StringType(), True),
    StructField("StoreID", StringType(), True),
    StructField("Product", StringType(), True),
    StructField("Quantity", IntegerType(), True),
    StructField("UnitPrice", DoubleType(), True),
    StructField("SalesAmount", DoubleType(), True),
    StructField("OrderTime", StringType(), True),          # can cast to Timestamp
    # later in Silver
    StructField("DeviceType", StringType(), True),
    StructField("Temperature", DoubleType(), True),
    StructField("DeviceStatus", StringType(), True),
    StructField("AnomalyFlag", IntegerType(), True),
    StructField("AnomalyType", StringType(), True)
])
# :white_check_mark: Paths (use subdirectories to avoid overlap with DLT system
files)
raw_path = "abfss://raw@vp07.dfs.core.windows.net/raw_data/"
schema_path = "abfss://raw@vp07.dfs.core.windows.net/dlt_schemas/bronze_orders_vp"
checkpoint_path =
"abfss://raw@vp07.dfs.core.windows.net/dlt_checkpoints/bronze_orders_vp"
@table(
    name="vp07_capstone.vp_schema.bronze_orders",
    comment="Bronze table - raw ingested IoT retail data via Auto Loader"
)
def bronze_orders():
    return (
        spark.readStream
            .format("cloudFiles")
            .option("cloudFiles.format", "csv")
```

```

        .option("header", True)
        .option("cloudFiles.schemaLocation", schema_path)
        .schema(iot_schema)    # :white_check_mark: manually set schema (no
infer needed)
        .load(raw_path)       # :white_check_mark: points to /raw_data/ folder
    )

@dlt.table(
    comment="Silver table - cleaned & typed data"
)
def silver_orders():
    df = dlt.read("vp07_capstone.vp_schema.bronze_orders")
    return (
        df.withColumn("Quantity", F.col("Quantity").cast("int"))
        .withColumn("UnitPrice", F.col("UnitPrice").cast("double"))
        .withColumn("OrderTime", F.to_timestamp("OrderTime")) # likely what you
want
    )
@dlt.table(
    comment="Gold table - sales aggregates for BI"
)
def gold_sales_summary():
    df = dlt.read("silver_orders") # :point_left: Just reference the DLT table
name
    return (
        df.groupBy("Region", "Product")
        .agg(
            F.sum(F.col("Quantity") * F.col("UnitPrice")).alias("TotalSales"),
            F.countDistinct("CustomerID").alias("UniqueCustomers"),
            F.avg("Quantity").alias("AvgQuantity")
        )
    )

```

```

from dlt import table, read
import pyspark.sql.functions as F
# =====
# Gold Layer Tables (capstone_retail.gold)
# =====
# 1. Daily / Monthly Sales per Region
@table(
    name="vp07_capstone.vp_schema.gold_sales_region",
    comment="Daily & Monthly Sales per Region"
)
def gold_sales_region():
    return (

```

```

        read("vp07_capstone.vp_schema.silver_orders")
        .withColumn("OrderDate", F.to_date("OrderTime", "M/d/yyyy H:mm"))
        .withColumn("YearMonth", F.date_format("OrderDate", "yyyy-MM"))
        .groupBy("Region", "OrderDate", "YearMonth")
        .agg(F.sum("SalesAmount").alias("TotalSales"))
    )
# 2. Device Anomaly Trend per Store
@table(
    name="vp07_capstone.vp_schema.gold_anomaly_trends",
    comment="Device Anomaly Trend per Store"
)
def gold_anomaly_trends():
    return (
        read("vp07_capstone.vp_schema.silver_orders")
        .withColumn("OrderDate", F.to_date("OrderTime", "M/d/yyyy H:mm"))
        .groupBy("StoreID", "OrderDate", "DeviceType", "AnomalyType")
        .agg(F.count("*").alias("AnomalyCount"))
    )
# 3. Conversion Rate Impact when devices fail
@table(
    name="vp07_capstone.vp_schema.gold_conversion_impact",
    comment="Conversion rate impact of device failures"
)
def gold_conversion_impact():
    df = read("vp07_capstone.vp_schema.silver_orders")
    total_orders = df.groupBy("StoreID").agg(F.count("*").alias("TotalOrders"))
    failed_orders = (
        df.filter(df.AnomalyFlag == 1)
        .groupBy("StoreID")
        .agg(F.count("*").alias("FailedOrders"))
    )
    return (
        total_orders.join(failed_orders, "StoreID", "left")
        .withColumn("FailedOrders", F.coalesce(F.col("FailedOrders"), F.lit(0)))
        .withColumn("FailureRate", F.col("FailedOrders") / F.col("TotalOrders"))
    )
# 4. Store Tier Classification
@table(
    name="vp07_capstone.vp_schema.gold_store_tiers",
    comment="Store tiers based on total sales (High / Medium / Low performers)"
)
def gold_store_tiers():
    df = (
        read("vp07_capstone.vp_schema.silver_orders")
        .groupBy("StoreID")
        .agg(F.sum("SalesAmount").alias("TotalSales"))
    )

```

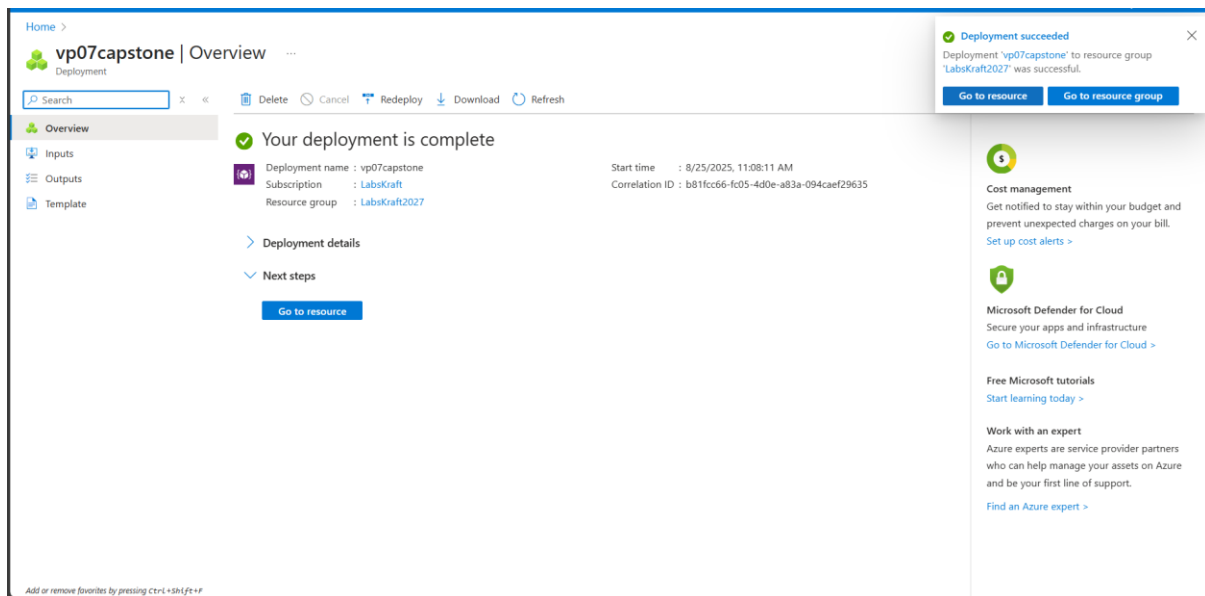
```

    )
    return (
        df.withColumn(
            "StoreTier",
            F.when(F.col("TotalSales") > 100000, "High")
              .when(F.col("TotalSales") > 50000, "Medium")
              .otherwise("Low")
        )
    )
)

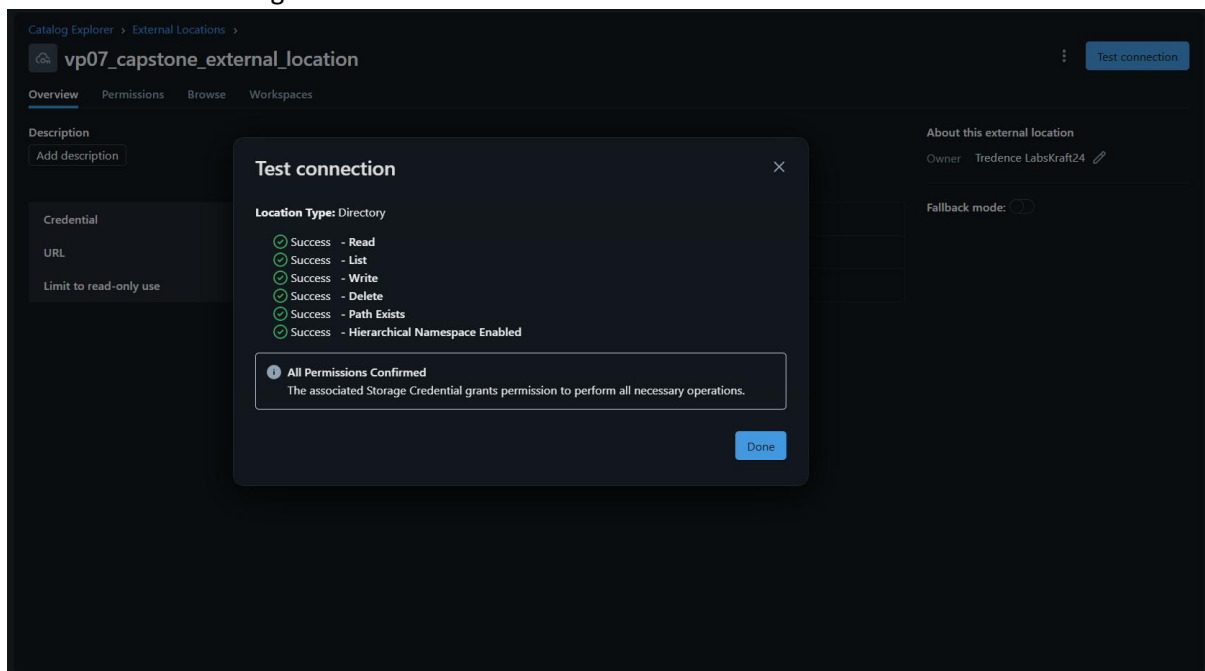
# 5. Weighted Average Sales vs Anomalies
@table(
    name="vp07_capstone.vp_schema.gold_weighted_sales_anomalies",
    comment="Weighted average of sales vs anomalies per store"
)
def gold_weighted_sales_anomalies():
    df = read("vp07_capstone.vp_schema.silver_orders")
    return (
        df.groupBy("StoreID")
        .agg(
            F.sum("SalesAmount").alias("TotalSales"),
            F.sum(F.when(F.col("AnomalyFlag") == 1,
1).otherwise(0)).alias("TotalAnomalies")
        )
        .withColumn("SalesPerAnomaly",
            F.when(F.col("TotalAnomalies") > 0, F.col("TotalSales") /
F.col("TotalAnomalies"))
            .otherwise(F.lit(None))
        )
    )
)

```


8. Create event hub



9. Create a new catalog's external location



Jobs & Pipelines >

vp07_final_capstone ☆ Send feedback

Development Production Settings Schedule Share Start

Run: 25/8/2025, 12:32:16 pm · Completed This update has already finished Show the latest update

Graph List

Streaming table

bronze_orders

Completed · 14s

1K 0

Pipeline details

Update details

Pipeline ID9c448bc5-9a0d-407f-bae4-39b005280b72

Pipeline typeETL pipeline

Source code/Users/tredence24@michaelkloudkrafthotmail.o
nmicrosoft.com/vp_new

Run asTredence LabsKraft24

TagsNone

Event log

Query history

All Info Warning Error Filter...

6 minutes ago

flow_progress

Flow 'vp07_capstone.vp_schema.bronze_orders' is RUNNING.

6 minutes ago

flow_progress

Completed a streaming update of 'vp07_capstone.vp_schema.bronze_orders'.

6 minutes ago

flow_progress

Flow 'vp07_capstone.vp_schema.bronze_orders' has COMPLETED.

6 minutes ago

update_progress

Update 3bffc4 is COMPLETED.

Jobs & Pipelines >

vp07_final_capstone ☆ Send feedback

Development Production Settings Schedule Share Start

Run: 25/8/2025, 12:48:05 pm · Completed Select tables for refresh

Graph List

Streaming table

bronze_orders

Completed · 3s

0 0

Materialized view

silver_orders

Completed · 8s

1K 0

Materialized view

gold_sales_summa...

Completed · 6s

28 0

Pipeline details

Update details

Pipeline ID9c448bc5-9a0d-407f-bae4-39b005280b72

Pipeline typeETL pipeline

Source code/Users/tredence24@michaelkloudkrafthotmail.o
nmicrosoft.com/vp_new

Run asTredence LabsKraft24

TagsNone

Event log

Query history

All Info Warning Error Filter...

1 minute ago

flow_progress

Flow 'vp07_capstone.vp_schema.gold_sales_summary' is RUNNING.

1 minute ago

dataset_life_cycle

Materialized View 'vp07_capstone.vp_schema.gold_sales_summary' has been created.

1 minute ago

flow_progress

Flow 'vp07_capstone.vp_schema.gold_sales_summary' has COMPLETED.

1 minute ago

update_progress

Update c90cf2 is COMPLETED.

11. Pipeline with bronze, silver and gold layer

Jobs & Pipelines > vp07_final_capstone ☆ Send feedback

Run: 25/8/2025, 2:02:12 pm · Completed · Select tables for refresh

Graph List

Pipeline details

Pipeline ID: 9c448bc5-9a0d-407f-bae4-39b005280b72

Pipeline type: ETL pipeline

Source code: /Users/tredence24@michaelkloudkrafthotmail.onmicrosoft.com/vp_new

Run as: Tredence LabsKraft24

Tags: None

Event log

Query history

All Info Warning Error Filter...

1 minute ago user_action User tredence24@michaelkloudkrafthotmail.onmicrosoft.com started an update.

1 minute ago create_update Update a13b87 started by USER_ACTION.

1 minute ago update_progress Update a13b87 is INITIALIZING.

1 minute ago unsupported_operation Magic commands (e.g. %py, %sql and ... with the exception of %pip within a Python notebook. Cells containing magic commands are ig...

12. PowerBI dashboard

